Panda joint identification: what was measured, how, and how it differs from franka::Model and Gaz et al. 2019

SE3PF, RoboHub Panda (FER), lab sessions of 2026-10-19 and 2026-10-20. Branch `ctrl-1-friction`, logs `logs/identify_1.csv` to `identify_3.csv` (identify_3 is the clean run used below).

1. Why

The paper's controller is feedback linearization, u = M(q) qdd* + C qd + G, with a PD on the tubular outputs. On the robot it stalled: the wrist joints did not move under commands of up to 0.06 N m, and every joint needed 0.5 to 1.5 N m before it moved at all. Two plant-model terms were suspect: the mass matrix franka::Model returns, and the joint friction, which the model does not have. Both were set by hand on the first day to get the arm moving; the identification below measures them, so that the runs with the paper's PD use a measured model and nothing tuned on the path-following runs themselves.

2. How it was measured (robot_driver <ip> identify)

The arm is held where it stands by the driver's damped joint hold (the safety stop's law, K_s = 300 N m/rad on joints 1 to 4, 50 and 20 on the wrist, clamped to 60 % of each torque limit). One joint at a time is released from the hold and given, in each direction:

(a) A torque ramp at 0.5 N m/s from zero (cap 2.5 N m on joints 1 to 4, 1.5 N m on the wrist) until the joint moves, defined as $|qd_j| \geq 0.02$ rad/s for 30 ms. The ramp's torque at the onset of motion is the raw breakaway. Because the net torque keeps growing after the true breakaway, a joint of inertia M reaches the threshold v_m only $\sqrt{(2\ M\ v_m\ /\ r)}$ later; the reported breakaway subtracts that lag, $\sqrt{(2\ M\ v_m\ r)}$, 0.1 to 0.2 N m here.

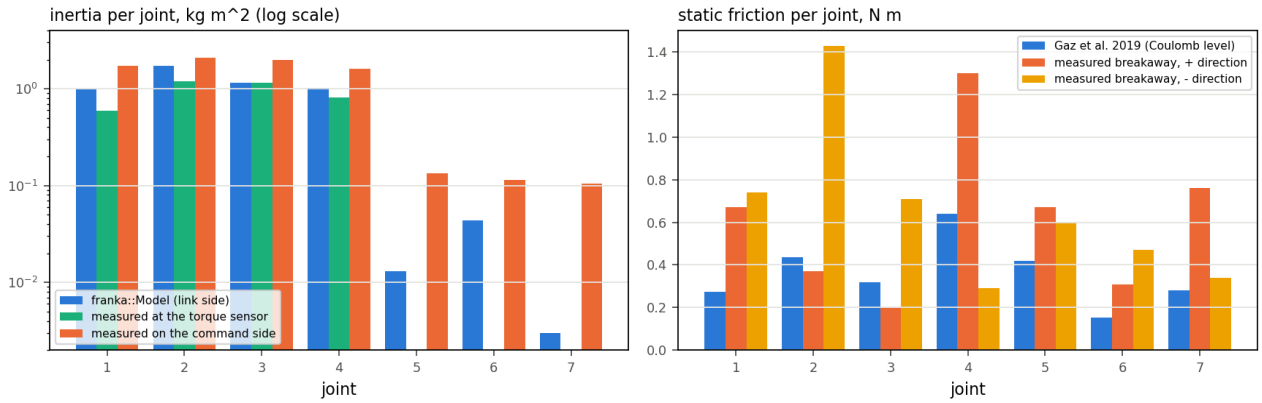
(b) Two torque steps above the breakaway, 1.0 and 1.5 N m on joints 1 to 4, 0.1 and 0.2 N m on the wrist, each held 0.3 s or until 0.08 rad of travel or 0.3 rad/s. The inertia follows from their difference without knowing the kinetic friction: $M = (\tau_2 - \tau_1) / (a_2 - a_1)$, with a the mean acceleration from motion onset to the end of the step. Done once with the commanded torque (what the controller must send) and once with the joint torque sensor τ_J minus gravity (what reaches the link). The kinetic friction is $\tau_1 - M\ a_1$.

Between pulses the hold settles for 2 s and re-latches the joint where the pulse left it; every change of command is blended over 0.15 s so the torque-rate margin holds. The run has the same gates as an experiment (log path, kinematics check, load confirmation, a static check of joint margins and the torque budget with the robot's gravity term, realtime, Idle, Enter), the monitor's joint, velocity, flange-speed and torque checks every cycle, and the safety stop on a violation. The whole sequence, seven joints, both directions, took 116 s; the driver fits the log itself and prints the table; `tools` has no separate script. Validation: a unit test on a simulated joint with Coulomb friction (fit within 0.05 N m and 15 %), and the mock robot with known friction and rotor inertia (within 0.08 N m and 20 %).

3. Results (identify_3, 2026-10-20, arm warm after an hour of motion)

joint	inertia, franka::Model (link side, kg m²)	inertia measured at the sensor (kg m²)	inertia measured on the command side (kg m²)	drive term to add to M (kg m²)	breakaway + / – (N m)	Gaz 2019 Coulomb (N m)
1	0.99	0.58 / 0.60	1.67 / 1.79	about 1.1	0.67 / 0.74	0.27
2	1.72	1.07 / 1.32	2.21 / 1.95	about 0.9	0.37 / 1.43	0.44
3	1.15	1.02 / 1.29	1.43 / 2.57	about 0.9 (noisy)	0.20 / 0.71	0.32
4	1.02	0.86 / 0.76	1.81 / 1.43	about 0.8	1.30 / 0.29	0.64
5	0.013	≈ 0	0.16 / 0.11	0.13	0.67 / 0.60	0.42
6	0.044	≈ 0	0.10 / 0.13	0.12	0.31 / 0.47	0.15
7	0.003	≈ 0	0.09 / 0.12	0.10	0.76 / 0.34	0.28

The model column is franka::Model's mass-matrix diagonal at the loops start configuration (read back from the hold run's log to 1e-5); the identification was run at a nearby posture (joint 7 half a turn away), so the sensor-side values match it only to the extent the posture matches. Gaz et al. 2019 values are $\phi_1/2$ of their sigmoid, the mean of its two asymptotes.



4. What it means

Inertia. The torque sensor sits after the gearbox and sees the link-side inertia, which agrees with franka::Model (0.6 to 1.3 kg m² on joints 1 to 4, about zero on the wrist). The command side sees that plus a drive term of about 1 kg m² on joints 1 to 4 and 0.10 to 0.15 on joints 5 to 7, consistent with the reflected rotor inertia through the 100:1 harmonic drives (10^{-4} and 10^{-5} kg m² rotors). franka::Model's M is therefore the wrong M for the torque command on every joint: on the wrist it under-sizes the torque per unit acceleration by a factor of 10 to 30, which is why the paper's controller could not move the wrist (0.02 to 0.06 N m inside the 30 % acceleration margin, against 0.3 to 0.8 N m of static friction), and on joints 1 to 4 by a factor of about 2.

Friction. Breakaway torques are 0.6 to 0.75 N m on joints 1 and 5, 0.3 to 0.5 on joint 6, and strongly direction dependent on the gravity-loaded joints 2 and 4 (0.4 against 1.4, 1.3 against 0.3 N m): the drive friction grows with the load torque it carries. Joint 3 (+) and joint 7 (–) came out low (0.20, 0.34) with sensor readings near zero and are not trusted. Against Gaz et al. 2019 the values are 1.5 to 3 times higher on most joints. That is not a contradiction: Gaz identified a kinetic (moving) friction from link-side torque-sensor data with no static term; these are static breakaways on the command side, which include the drive's stiction and what the robot's inner torque loop fails to deliver (the sensor reads only 30 to 60 % of a command while a joint is stuck or slipping). Static friction of a harmonic drive is typically 1.5 to 3 times the kinetic value. Run-to-run scatter is 20 to 40 % on friction (joint 1: 0.45/0.52 in run 2, 0.67/0.74 in run 3) and about 15 % on inertia; the first run of the day, with the arm cold, gave the lowest breakaways on the wrist.

5. What is in the controller now

Experiment `loops-friction`: the paper's PD (poles at –6, k_t = 15, no integral action) with `rotorInertia = (1.0, 1.0, 1.0, 1.0, 0.13, 0.12, 0.10)` kg m² added to franka::Model's M in the command, and a symmetric Coulomb feedforward along the reference joint velocity at the smaller of the two breakaways per joint, `Fc = (0.65, 0.40, 0.20, 0.30, 0.60, 0.30, 0.35)` N m. Both are plant-model terms of eq:fb; the control law is unchanged. `loops-inertia` is the same without the friction term. The direction dependence of joints 2 and 4 is not in the model and is the main remaining model error, which is what the PD's steady errors will reflect.

6. Caveats and next steps

One posture, two runs; the fits are from 0.3 s steps at 0.6 to 1.5 rad/s² (joints 1 to 4) and 0.6 to 4 rad/s² (wrist). For the paper: repeat the identification at two or three postures along the path and warm, report mean and spread; consider a direction-dependent or load-dependent friction term for joints 2 and 4; and run the perturbation experiments (payload mass 30 % wrong in Desk, friction model scaled by 0.5 and 1.5) with the measured model frozen, which is the robustness demonstration the paper's argument needs.

Files: `include/se3pf/identify.hpp`, `src/identify.cpp`, the `identify` command in `apps/robot/robot_driver.cpp`, `tests/test_identify.cpp`, the SIL test in `tests/sil/test_robot_driver.cpp`; logs `logs/identify_1..3.csv` (columns: t, phase, joint, dir, pulse, pulse_tau, tau, q, dq, tauJ, G).